Hello Logan, and the team at Automwrite,

Please find my solution to the given cto-take-home assignment.

I have committed it to my GitHub account - https://github.com/jay-jayasimha/take-home.git (my_changes branch)

I have analyzed userintent.txt and userintent-2.txt with Python spaCy NLP tools to return a JSON object that highlighted 4 key entities:

1. Intent {transfer}
2. Source (from plan)
3. Destination (to plan)
4. portfolio Profile (Aggressive, Moderate, Low, suitable risk growth plans)

I developed custom python models (extending Spacy's models) on top of:

NAME	SPACY	VERSION
en_core_web_trf	>=3.8.0,<3.9.0	3.8.0
en_core_web_md	>=3.8.0,<3.9.0	3.8.0
en_core_web_sm	>=3.8.0,<3.9.0	3.8.0
en_core_web_lg	>=3.8.0,<3.9.0	3.8.0

I had enhanced the ORG, Intent, and portfolio models to recognize the inputs on top of core models such as the en_core_web_trf and _lg models.

I also used OpenAI Nature Language Processing intent analysis to analyze the input user intent. I wanted to use the OpenAI web service for this but ran into some issues with security, calling from Java web service.

I used the following curl inputs to test:

```
url -X POST \
  http://localhost:8080/api/user-request \
  -H "Content-Type: multipart/form-data" \
  -F file=@src/main/resources/userintent/userintent.txt
```

And

```
url -X POST \
  http://localhost:8080/api/user-request \
  -H "Content-Type: multipart/form-data" \
  -F file=@src/main/resources/userintent/userintent-2.txt
```

To generate the recommendation.docx. I have modified the template doc a bit and created recommendation_template_Jay.docx that you will see in the resources/templates folder.

Due to time limitations, I was not able to spend enough time to either 1. Call the python tools from Java and 2. Call the OpenAI web services from Java. I wanted to send the solution out today.

You will find additional code that I had placed there for debugging and testing of the model objects such as client and organization.

Thank you very much for providing an exciting take home project to complete and enjoyed working on it.

Thanks, and with very best wishes,
Jay